Topic 1.8—The Squeeze Theorem

1. Why do our limit properties not work to evaluate $\lim_{x\to 0} x^2 \sin\frac{1}{x}$?

   a. What two values is $\sin x$ sandwiched between? Fill in the inequality.

   $$_____ \leq \sin x \leq _____$$

   b. What about $\sin\frac{1}{x}$? Has the range of the function changed?

   $$_____ \leq \sin\frac{1}{x} \leq _____$$

   c. What would happen if we applied a vertical stretch or shrink? Has the range of the function changed?

   $$_____ \leq 3\sin\frac{1}{x} \leq _____ \qquad _____ \leq \frac{1}{2}\sin\frac{1}{x} \leq _____$$

   $$_____ \leq k\sin\frac{1}{x} \leq _____$$

   d. Using the ideas from 3d, what would $x^2 \sin\frac{1}{x}$ be sandwiched between? (Hint: they will be functions!)

   $$_____ \leq x^2\sin\frac{1}{x} \leq _____$$

2. Now, we can start talking about limits!

   a. What is the limit as *x* approaches 0 of your lower-bound function?

   b. What is the limit as x approaches 0 of your upper-bound function?

   c. What is $\lim_{x\to 0} x^2 \sin\frac{1}{x}$?

Important Ideas:

Check Your Understanding!

1. Let $k(x) = -x^2 - 2x + 3$, and $m(x) = \frac{1}{2}x^2 + x + \frac{13}{2}$. If f is a function that satisfies $k(x) \leq f(x) \leq m(x)$, explain why the squeeze theorem can NOT be used to find $\lim_{x \to -1} f(x)$.

2. Selected values of two continuous functions, f and g, are given in the table below.

x	$f(x)$	$g(x)$
-3	9	15
0	7	8
1	11	11
2	-1	2
7	-5	-4

If $f(x) \leq k(x) \leq g(x)$ for all values of x, find $\lim_{x \to 1} k(x)$. Explain your reasoning.